



THE UNIVERSITY OF
MELBOURNE

SWEN90010 - High Integrity Systems Engineering

Lecture 1: Introduction

Toby Murray



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Level 2, Melbourne Connect



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@tobycmurray



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ADMINISTRIVIA

Lectures

Wednesday 10am
Friday 12pm



Toby Murray



Christine Rizkallah

firstname.lastname@unimelb.edu.au

Workshops



in-person

Head Tutor

Louis Cheung

cheung.l@unimelb.edu.au

**Workshops start Tuesday
of Week 2**

Alex Epstein

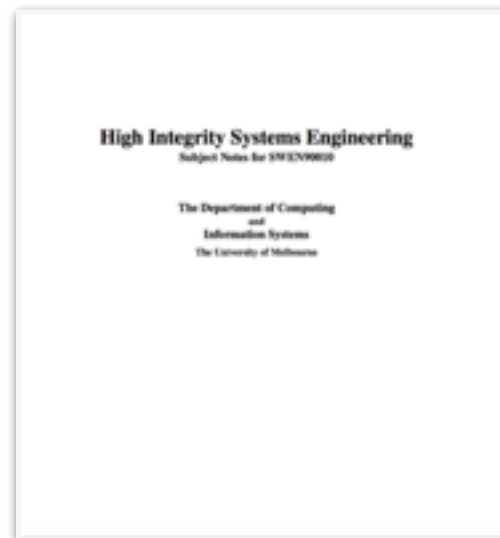
epsteina@unimelb.edu.au

Jin Deng

jincheng.deng.1@unimelb.edu.au

Subject Notes (on LMS)

Booklet:



A living resource.
Will be
updated as we go.

Lecture Slides



Assignments



3 Assignments, comprise 50% of your grade for the subject

Hurdle: 25/50 for your assignments overall

(**Exam Hurdle:** 25/50 for the written exam)

Rough Timeline

(might change slightly as we go)

1. 10 marks, released Week 2, due Monday Week 5
2. 20 marks, released Week 5, due Friday Week 7
3. 20 marks, released Week 8, due Friday Week 11

Group Work

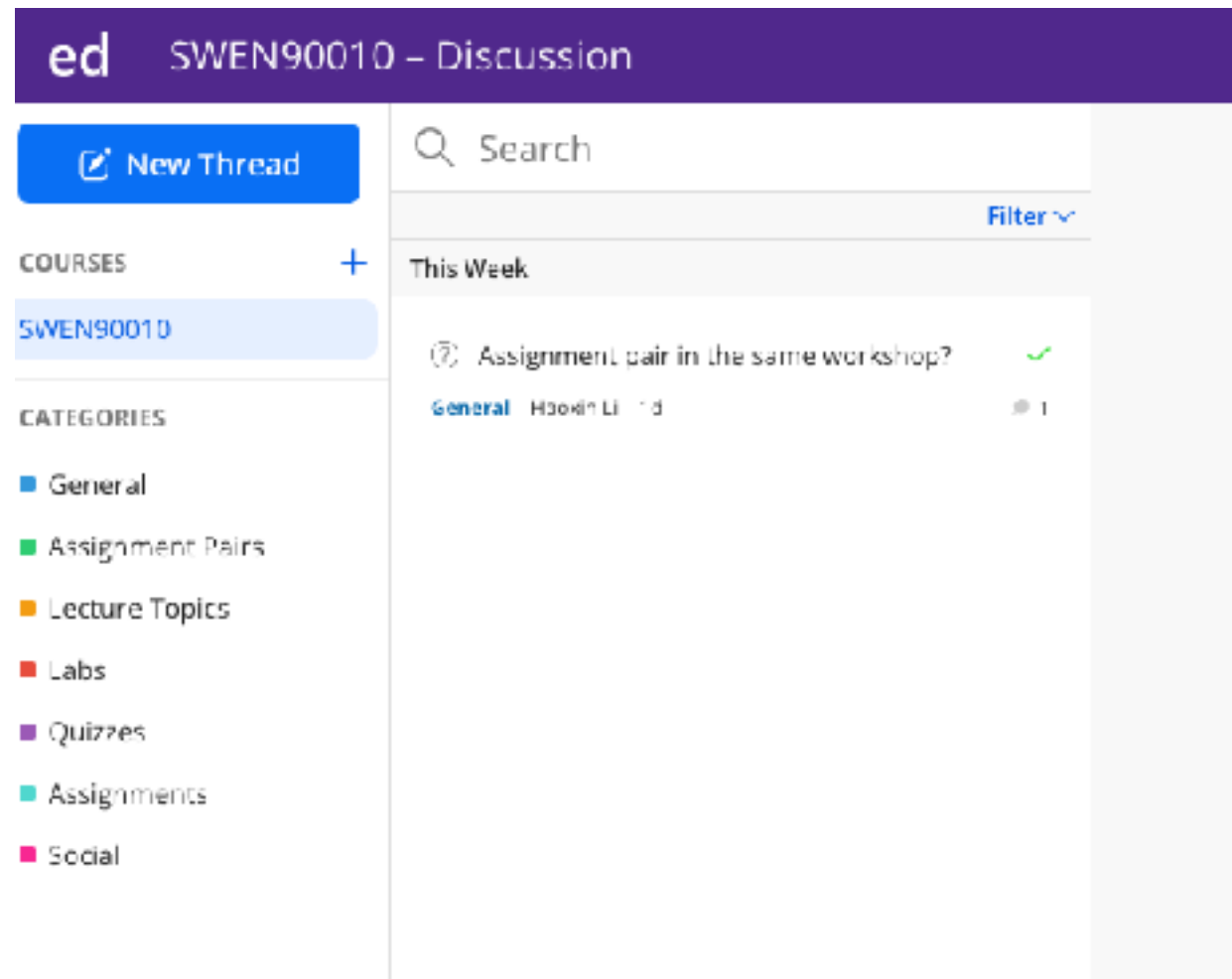
Assignments you will do in self-selected pairs

This week: choose your pair record it
on the LMS:

- 1 Go to **Canvas > People > Assignments Pair**
- 2 Click “Join” next to an available group (each pair has 2 members max)
- 3 If no group is available, you can create a new group and invite a partner
- 4 Confirm your selection before the deadline

On Wednesday Week 2 we will allocate any remaining
students

Discussion Board



Don't reveal information about your own assignment solutions here.

Do answer others' questions.

Academic Integrity



<https://academichonesty.unimelb.edu.au/>

The University misconduct policy applies.

Students are encouraged to discuss the assignment topics, but all submitted work must represent the pair's understanding of the topic.

We take academic misconduct very seriously. In this subject in the past, we have successfully prosecuted several students that have breached the university policy. Often this results in receiving 0 marks for an assignment, and in some cases has resulted in failing the subject.

Academic Integrity



The University of Melbourne's Student Academic Integrity Policy (MPF1310) makes clear that all work submitted by an individual student must be their own. In the case of group work, the individual contribution of each student must be their own work.

FEIT AI Use Policy

FEIT allows the use of Generative AI (GenAI) in open assessment tasks, unless the assessment instructions include a specific "AI Use Statement" that limits or prohibits the use. Using GenAI is not an academic integrity breach on its own. However, your submitted work must still be your own, and you remain fully responsible for meeting all academic integrity requirements – this includes properly acknowledging and referencing any GenAI you use. We encourage all students to take shared responsibility for their learning, and consider potential impacts on secure assessment tasks.

If anything in this policy is unclear, please speak with your teaching team for guidance.



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HIGH INTEGRITY SYSTEMS ENGINEERING

Engineering vs Programming



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OpenClaw

THE AI THAT ACTUALLY DOES THINGS.

Clears your inbox, sends emails, manages your calendar, checks you in for flights.

All from WhatsApp, Telegram, or any chat app you already use.

Engineering in the Age of AI



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March 3, 2026 | 5 min read

 Add Us On Google 

AI was supposed to save coders time. It may be doing the opposite

Studies find AI helps developers release more software—while logging longer hours and fixing problems after the code goes live

BY STEVEN MELENDEZ EDITED BY ERIC SULLIVAN

Watershed Moment for AI–Human Collaboration in Math › Twenty-first-century Fields Medal proof formalized for the first time

BY BENJAMIN SKUSE

02 MAR 2026

Feb 28 2026

Leonardo de Moura [home](#) [about me](#) [publications](#) [blog](#) [slides](#) [research](#)

When AI Writes the World's Software, Who Verifies It?

AI Is Rewriting the World's Software

Engineering

Methods for making large-scale software that is **fit for purpose**

Criteria

Security Correctness

Performance Reliability

Usability Scalability

Time to market

Development cost

Methods

Requirements analysis

Design Testing

Reviews
(code, design, requirements)

Prototyping Formal
Analysis

High Integrity Systems



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Is Knight's \$440 million glitch the costliest computer bug ever?

2012

Yes, pac

By Post Staff Report

Airbus issues software bug alert after fatal plane crash

May 2015

What to learn from the \$10 million Subway POS hack

Toyota's killer firmware: Bad design and its consequences

Dunn Michael -October 28, 2013

Th **Was**

The day a software bug almost killed the Spirit rover

2013

High Integrity Systems

Ones in which **failure** has unacceptable consequences

Loss of life

Loss of money

Environmental damage

Loss of mission

Loss of privacy

A Thought Experiment

Would you fly on a plane whose software
was written by a vibe coder?

Why do you trust flight software?

What do those engineers do differently?

What this subject is about.

Safety Critical Systems

Obvious ones:

Network-enabled Pacemakers

Software in Car ECUs

Railway Signalling

Less obvious ones:

Ambulance dispatch system

Security Critical Systems



Almost everything, as **incentives** improve for attackers
as more **things** of value become **network accessible**

Computer systems at your GP's clinic or hospital

KIM ZETTER SECURITY 03.30.16 1:31 PM

WHY HOSPITALS ARE THE PERFECT TARGETS FOR

**Sh... IoT just got real: Mirai botnet attacks
targeting multiple ISPs**

Now ZyXEL and D-Link routers from Post Office and TalkTalk
under siege

NAS for AV recordings for Court Services Victoria

Finance Critical Systems



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The Cost of Downtime

by **Andrew Lerner** | July 16, 2014 | [90 Comments](#)

This is a follow-up to last weeks post on [network downtime](#) as several have asked “*What is the cost of network downtime?*” Based on industry surveys, the number we typically cite is \$5,600 p/minute, which extrapolates to well over \$300K p/hour (see [Ensure Cost Balances Out With Risk in High-Availability Data Centers](#) by Dave Cappuccio).

However, this is just an **average** and there is a large degree of variance, based on the characteristics of your business and environment (i.e., your vertical, risk tolerance etc). For example, this [Avaya study](#) indicates the range is from \$140K to \$540K p/hour. That said, why rely

Hosting providers

Mission Critical

Think NASA

Cost of NASA's Next Mars Rover Hits Nearly \$2.5 Billion

By Amy Svitak, Space News Staff Writer | February 3, 2011 07:00pm ET

High Integrity Engineering



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Old London Bridge, 1632



London Bridge today

Safety and Security Analysis



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Handbook

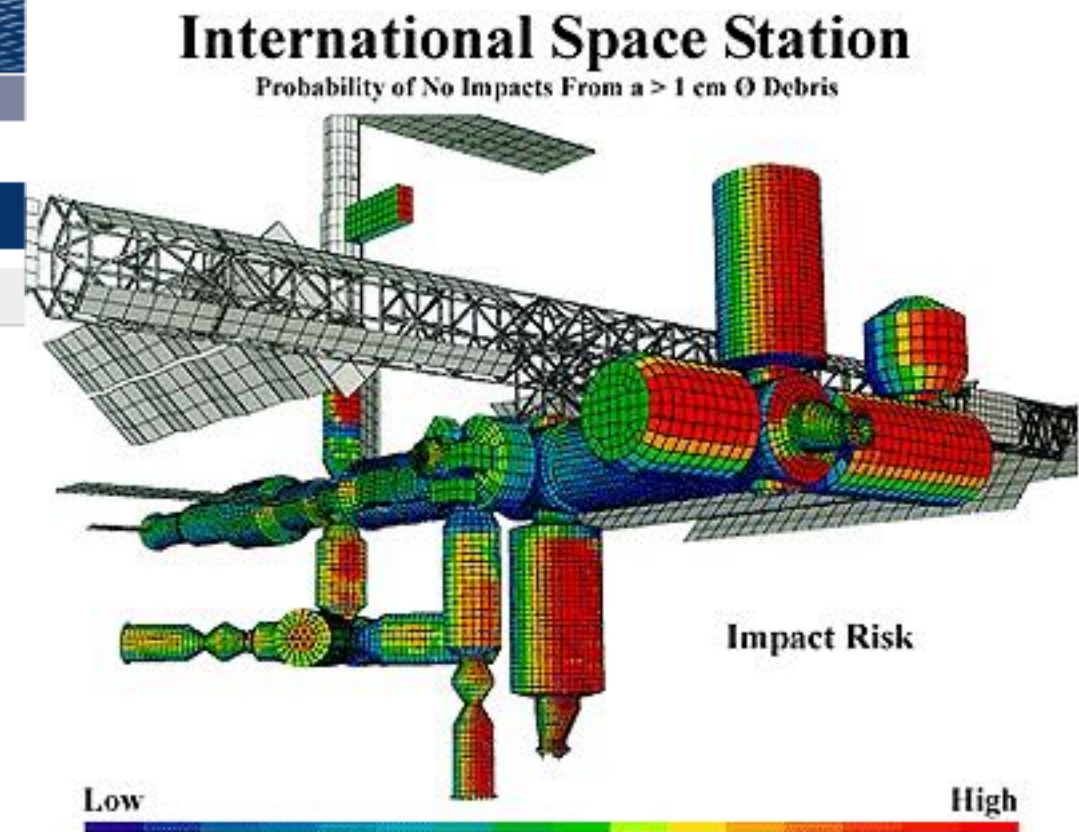
Links: [University](#)

[Handbook](#) > [CVEN30008 Engineering Risk Analysis](#)

CVEN30008 Engineering Risk Analysis

On this page: | [Subject Overview](#) | [Breadth options](#) | [Related Program\(s\)](#)

Credit Points:	12.5												
Level:	3 (Undergraduate)												
Dates & Locations:	This subject has the following teaching availabilities in 2017: Semester 1, Parkville - Taught on campus. Show/hide details <table><tr><td>Pre-teaching Period Start</td><td>not applicable</td></tr><tr><td>Teaching Period</td><td>27-Feb-2017 to 28-May-2017</td></tr><tr><td>Assessment Period End</td><td>23-Jun-2017</td></tr><tr><td>Last date to Self-Enrol</td><td>10-Mar-2017</td></tr><tr><td>Census Date</td><td>31-Mar-2017</td></tr><tr><td>Last date to Withdraw without fail</td><td>05-May-2017</td></tr></table> Timetable can be viewed here. For information about these dates, click here. Contact Hours: 36 hours (Lectures: 2 hours per week; Tutorials: 1 hour per week)	Pre-teaching Period Start	not applicable	Teaching Period	27-Feb-2017 to 28-May-2017	Assessment Period End	23-Jun-2017	Last date to Self-Enrol	10-Mar-2017	Census Date	31-Mar-2017	Last date to Withdraw without fail	05-May-2017
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Time Commitment:	Total Time Commitment: 170 hours												



Hazard analysis and fault prediction

Formal Methods

Engineers in established fields use applied mathematics to predict the behavior and properties of their designs with great accuracy. Software engineers, despite the fact that their creations exhibit far more complexity than physical systems, do not generally do this . . .

[T]he applied mathematics of software is formal logic, and calculating the behavior of software is an exercise in theorem proving. Just as engineers in other disciplines need the speed and accuracy of computers to help them perform their engineering calculations, so software engineers can use the speed and accuracy of computers to help them prove the . . . theorems required to predict the behavior of software.

John Rushby. *Formal methods and critical systems in the real world*

Formal Methods

Software Engineering

Requirements

Design

Implementation



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Formal Methods

Properties

Alloy

Design Model

Hoare Logic

SPARK

Code Semantics

Small trustworthy foundation

microkernel with verified **functional correctness**, plus verified **integrity** and **confidentiality** enforcement

High assurance components in presence of other components

seL4 API:

- IPC
- Threads
- VM
- IRQ
- Capabilities

Untrusted

Legacy Apps

Linux Server

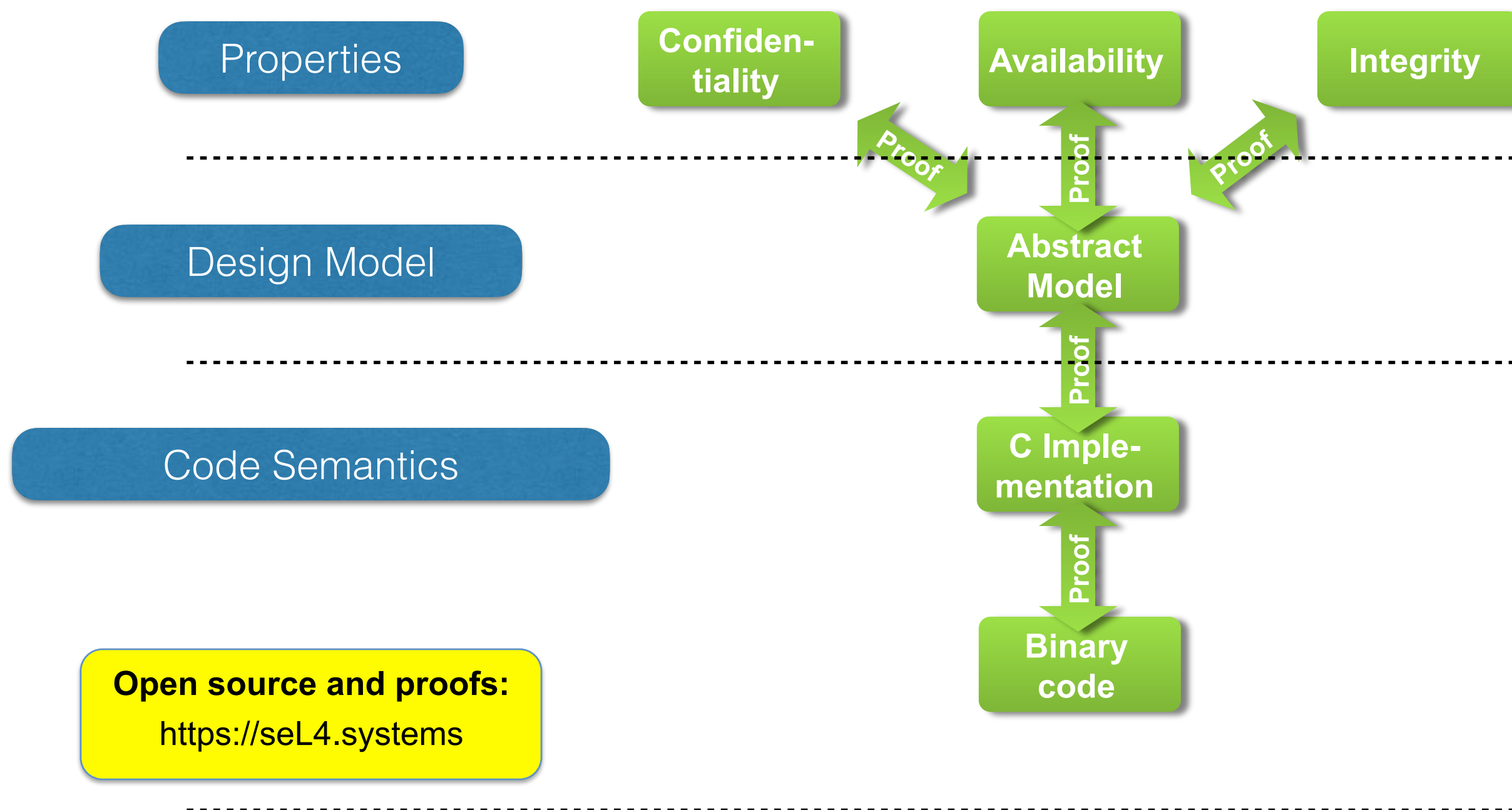
Trusted

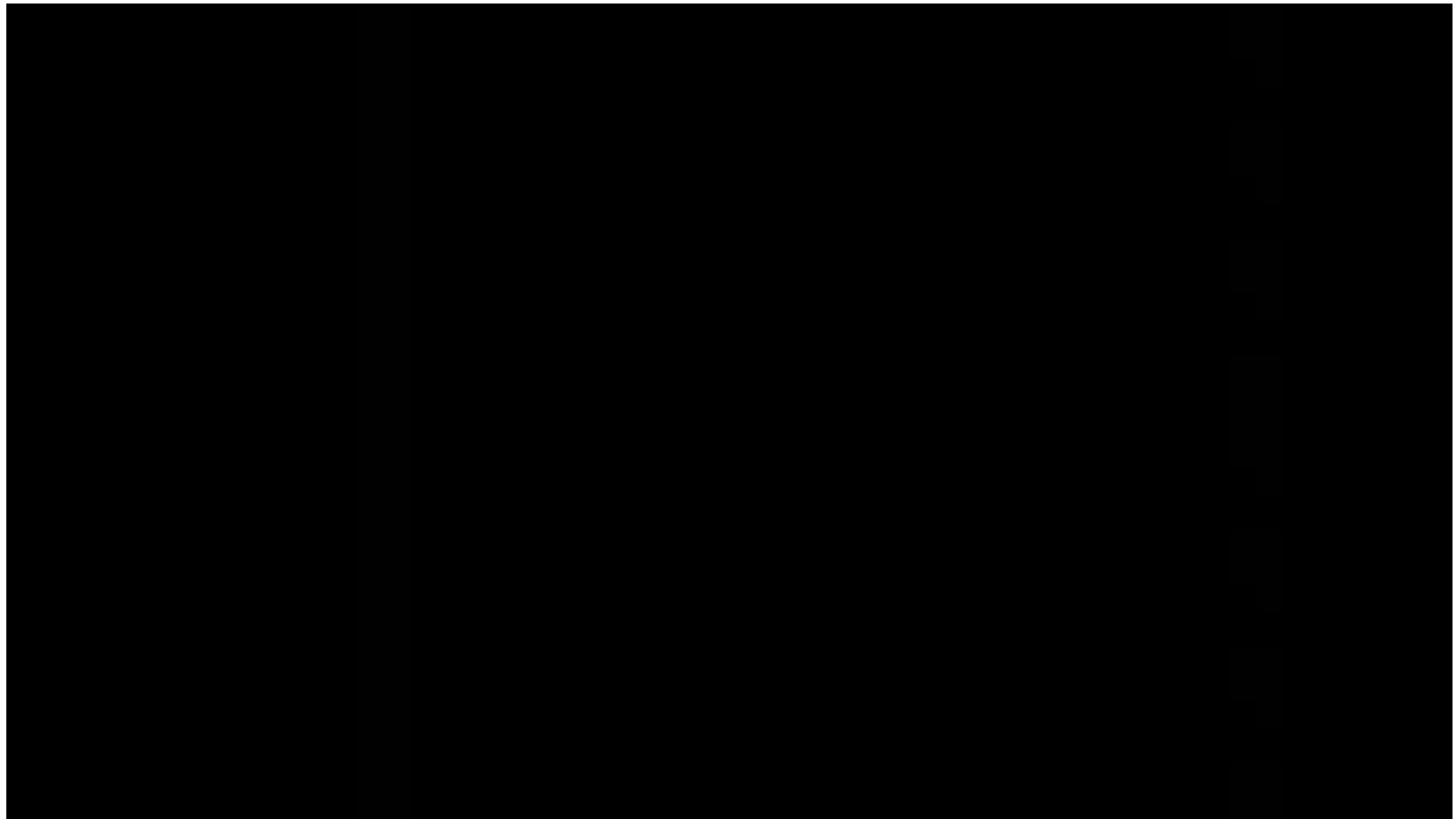
Sensitive App

Trusted Service

seL4

Hardware





 **sel4** Inside!



Case 14 Inside!

Cost of Formal Methods



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Toby Murray @tobycmurray.bsky.social · 13h

Almost no matter what base of human effort this was situated on, 200K lines of proof in 2 weeks is remarkable. In Isabelle/HOL we used to reckon on about 10K lines of proof = 1 person-year.

eaper

Properties

Make predictions
about system
behaviour

Reason about
low-level attacks

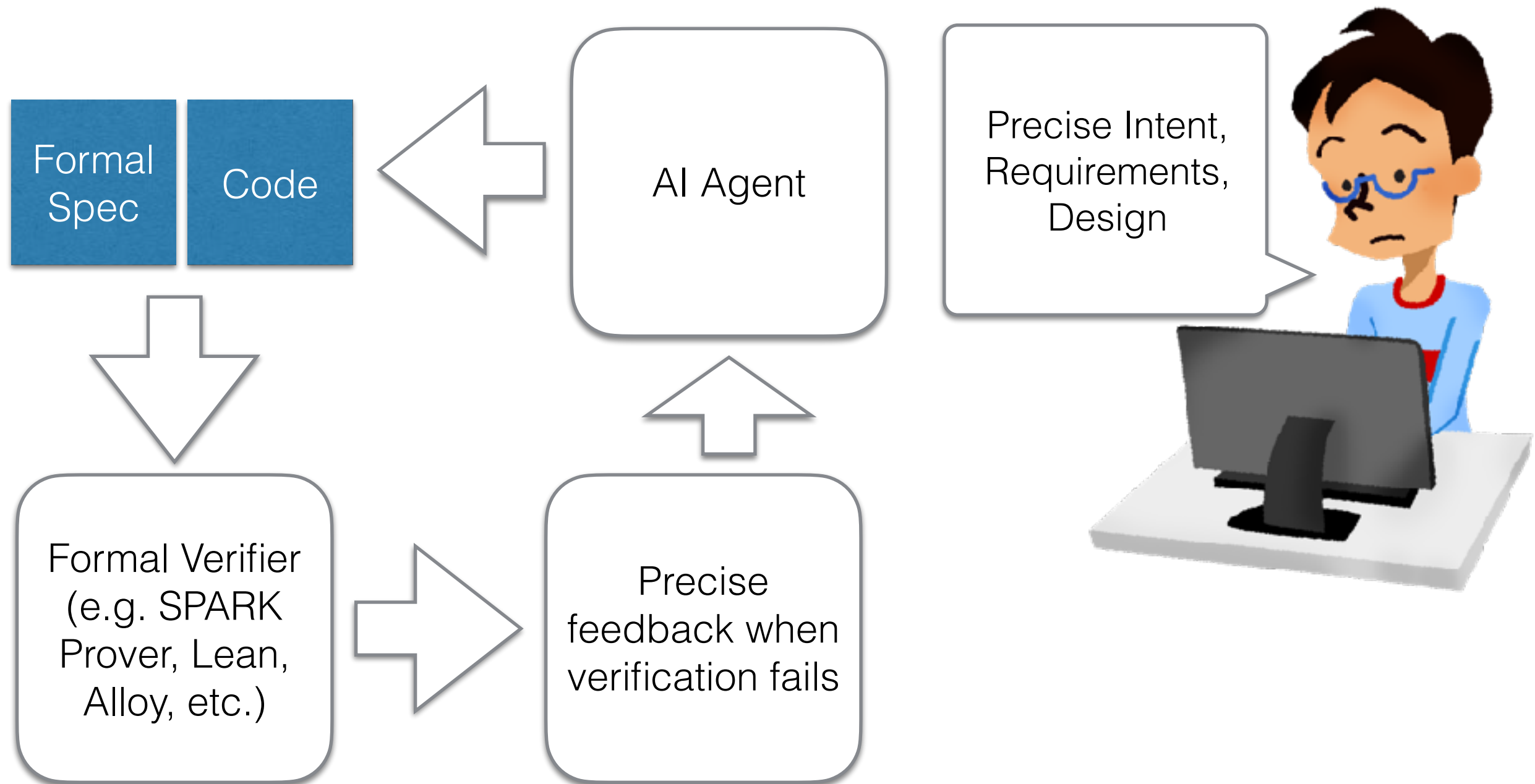
Design Model

Code Semantics

```
roshan@linuxmint: ~$ objdump -M intel -D | grep -A20 main:
00401040: 55          push    ebp
00401041: 89 e5       mov     ebp,esp
00401042: 83 e4 f0    and     esp,0xfffff0
00401043: 83 ec 20    sub     esp,0x20
00401044: c7 46 24 1c 00 00 00 mov     DWORD PTR [esp+0x1c],0x0
00401045: 00         nop
00401046: 4b 11       jnp     00401047 <main+0x14>
00401047: c7 04 24 b8 04 00 mov     DWORD PTR [esp+0x1c],0x0
00401048: 48 fd ff ff call    00401049 <main+0x15>
00401049: 83 44 24 1c 00 00 mov     DWORD PTR [esp+0x1c],0x0
0040104a: 83 7c 24 1c 00 00 cmp     DWORD PTR [esp+0x1c],0x0
0040104b: 7e e8       jle     0040104c <main+0x12>
0040104c: 90 90 00 00 00 00 mov     eax,0x0
0040104d: c9         leave  eax
0040104e: c3         ret
0040104f: 90         nop
00401050: 90         nop
00401051: 90         nop
00401052: 90         nop
00401053: 90         nop
```

More Expensive

Engineering with AI?



Is it worth it?

For high-assurance domains, definitely.
(space, aviation, Defence, etc.)

In these domains there is no cheaper way
to develop software.

For others, AI is changing the incentives to make formal
methods worthwhile

Commercial Formal Methods



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alTRan

| galois |

Course Outline



Loosely based around Altran's development model

But not using all of their toolset

Safety/Security Analysis

Design Modelling and Analysis (Alloy)

Ada

Proof (Hoare Logic) & Safe Languages (SPARK)

Fault Tolerant Design